

STATISTIC

THE POISSON DISTRIBUTION

The poisson random variable is the total number of occurrences of discrete events within a given time or space.

These conditions must be met:

1. events occur independently
2. events cannot occur simultaneously
3. For a given interval (of time or space)
 - the number of events occurring is proportional to the size of the interval i.e. events occur at a constant rate.

• This rate is called λ (lambda) "lamna"

• λ is also the mean of the distribution

• Notation if X has a poisson distribution with mean = λ then

$$X \sim P_0(\lambda)$$

Poisson Formula:

If $X \sim P_0(\lambda)$

$$P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}$$

where $X = \{0, 1, 2, 3, \dots, r, \dots\}$

Eg. A pacific Island gets on average 7.2 cyclones per year.
Assuming a poisson distribution
Find probability of

- (a) 5 cyclones next year
- (b) No cyclones next year
- (c) More than 2 cyclones next year
- (d) 1 cyclone in the next 3 months.

ANS:

$$X \sim P_o (\lambda = 7.2 \text{ per year})$$

$$(a) \quad r = 5$$
$$P(X = 5) = \frac{e^{-7.2} \times 7.2^5}{5!} = 0.120 \quad (3sf)$$

$$(b) \quad r = 0$$
$$P(X = 0) = \frac{e^{-7.2} \times 7.2^0}{0!} = 7.47 \times 10^{-4} \quad (3sf)$$
$$= 0.000747$$

$$(c) \quad r > 2$$

$$P(X > 2) = 1 - P(X = 0, 1, 2)$$
$$= 1 - \left[\frac{e^{-7.2} \times 7.2^0}{0!} + \frac{e^{-7.2} \times 7.2^1}{1!} + \frac{e^{-7.2} \times 7.2^2}{2!} \right]$$
$$= 1 - [0.025473 \dots]$$
$$= 0.97452 \dots$$
$$= 0.975 \quad (3sf)$$

$$(d) \quad 3 \text{ months} = \frac{1}{4} \text{ of a year} \quad \lambda = 7.2 \times \frac{1}{4} = 1.8$$

$$r = 1$$
$$P(X = 1) = \frac{e^{-1.8} \times 1.8^1}{1!} = 0.29753 \dots = 0.298 \quad (3sf)$$

CUMULATIVE POISSON FORMULA

$$= \frac{e^{-\lambda} \lambda^0}{0!} + \frac{e^{-\lambda} \lambda^1}{1!} + \frac{e^{-\lambda} \lambda^2}{2!} + \dots + \frac{e^{-\lambda} \lambda^n}{n!}$$

$$P(X \leq n) = e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2} + \frac{\lambda^3}{3!} + \dots + \frac{\lambda^n}{n!} \right]$$

$$\text{e.g. } P(X \leq 3) = e^{-3.2} \left[1 + 3.2 + \frac{3.2^2}{2} + \frac{3.2^3}{6} \right]$$

$$(\lambda = 3.2) = 0.603 \text{ (3sf)}$$

Do calc. R to L

SUMS OF POISSON RANDOM VARIABLES (RVs)

The sum of 2 or more ^{independent} poisson variables will also have a poisson distribution

and

The lambda of the sum equals to the sum of the lambdas

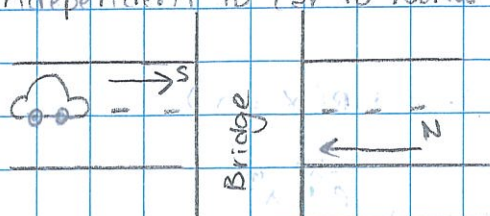
$$X \sim P_0(3.2)$$

$$Y \sim P_0(1.8)$$

then $(X+Y) \sim P_0(5.0)$

$$L \quad 3.2 + 1.8$$

cars going to south are independent to car to north



X = NO. of cars going to south $\lambda_x = 2.1$ per min

Y = NO of cars going to north $\lambda_y = 3.6$ per min

What is the probability that in the next minute 5 cars pass under the bridge?

Let $S = X + Y$

$$\lambda_s = 2.1 + 3.6 = 5.7$$

$$P(S=5) = \frac{e^{-5.7} \times 5.7^5}{5!} = 0.168$$

USING POISSON TO APPROXIMATE THE BINOMIAL

If $X \sim B(n, p)$ i.e. X is Binomial then the poisson formula can be used to give approximate probabilities provided: n is large [$n \geq 50$]

p is small

and $np < 5$

$$B(n, p) = {}^n C_x P^x (1-p)^{n-x}$$

e.g. $X \sim B(200, 0.02)$ Find $P(X=2)$ using

(a) Binomial expansion formula

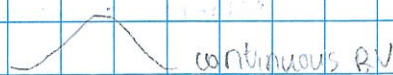
(b) Poisson formula $\lambda = np$

$$(a) P(X=2) = {}^{200} C_2 (0.02)^2 (0.98)^{198} = 0.1457726 \approx 0.146$$

$$(b) \lambda = np = 200 \times 0.02 = 4$$

$$P(X=2) = \frac{e^{-4} \times 4^2}{2!} = 0.146525 \approx 0.147$$

the bigger the λ the better the normal can satisfy normal curve



State the distribution

- B give n and p
- P_o give λ
- N give μ and σ^2

Variance σ^2

THE MEAN & VARIANCE OF A POISSON R.V

- In theory, if $X \sim P_o(\lambda)$

then mean of $X = E(X) = \lambda$

and variance of $X = \text{Var}(X) = \lambda$

- conversely, if data does not have mean = variance,
it cannot be POISSON

$$\mu = \text{var}$$

THE NORMAL APPROXIMATION TO THE POISSON

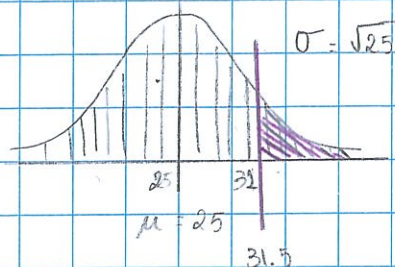
if $X \sim P_o(\lambda)$ and $\lambda \geq 15$ (ie X is poisson),
the poisson probability can be found (approximately) by
using a normal distribution with mean = λ

$$\text{std. dev} = \sqrt{\lambda}$$

NOTE: use continuity correction

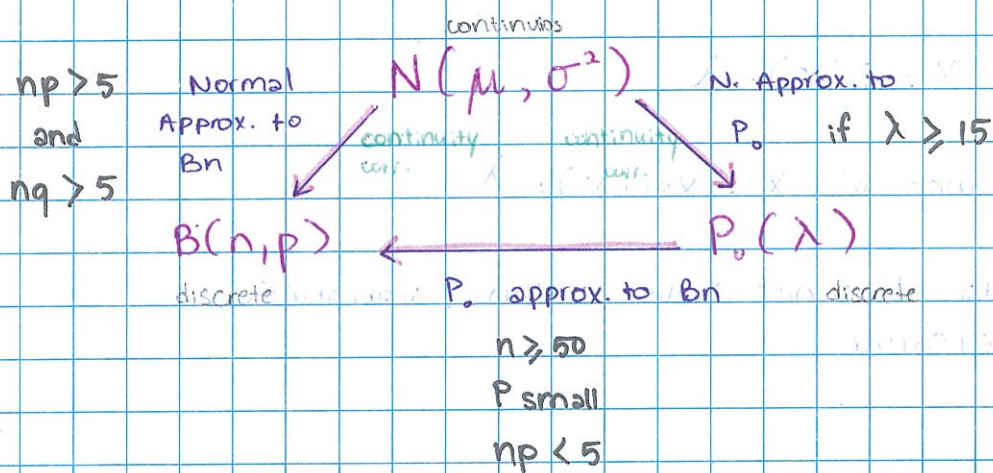
31.5 rather than 31

Eg The annual number of deaths nationally from a disease can be modelled by poisson $X \sim P_o(\lambda = 25)$
 what is prob that next year more than 31 people die?



$$\begin{aligned} P(X > 31) \\ &= P(X > 31.5) \\ &= P\left(Z > \frac{31.5 - 25}{\sqrt{25}}\right) \\ &= P(Z > 1.3) \\ &= 0.0968 \end{aligned}$$

Summary of Approximation



$\mu = 25$

$P(X > 31) = 31.5$

$P(X \geq 31) = 30.5$

$P(X < 31) = 30.5$

$P(X \leq 31) = 31.5$